

MA3259 Lecture 1

Introduction to Sequence Comparison

LX Zhang

Department of Mathematics
National University of Singapore
matzlx@nus.edu.sg

Objective. Develop competitive working knowledge in formulating biological problems in computational terms and solving them using the algorithmic and math approach.

-- See what everyone else has seen,
but think them mathematically.

Assessment mode:

- a) Tutorial attendance and discussion (5%);
- b) Three assignments (15 %);
- c) (open book) final examination (80 %).

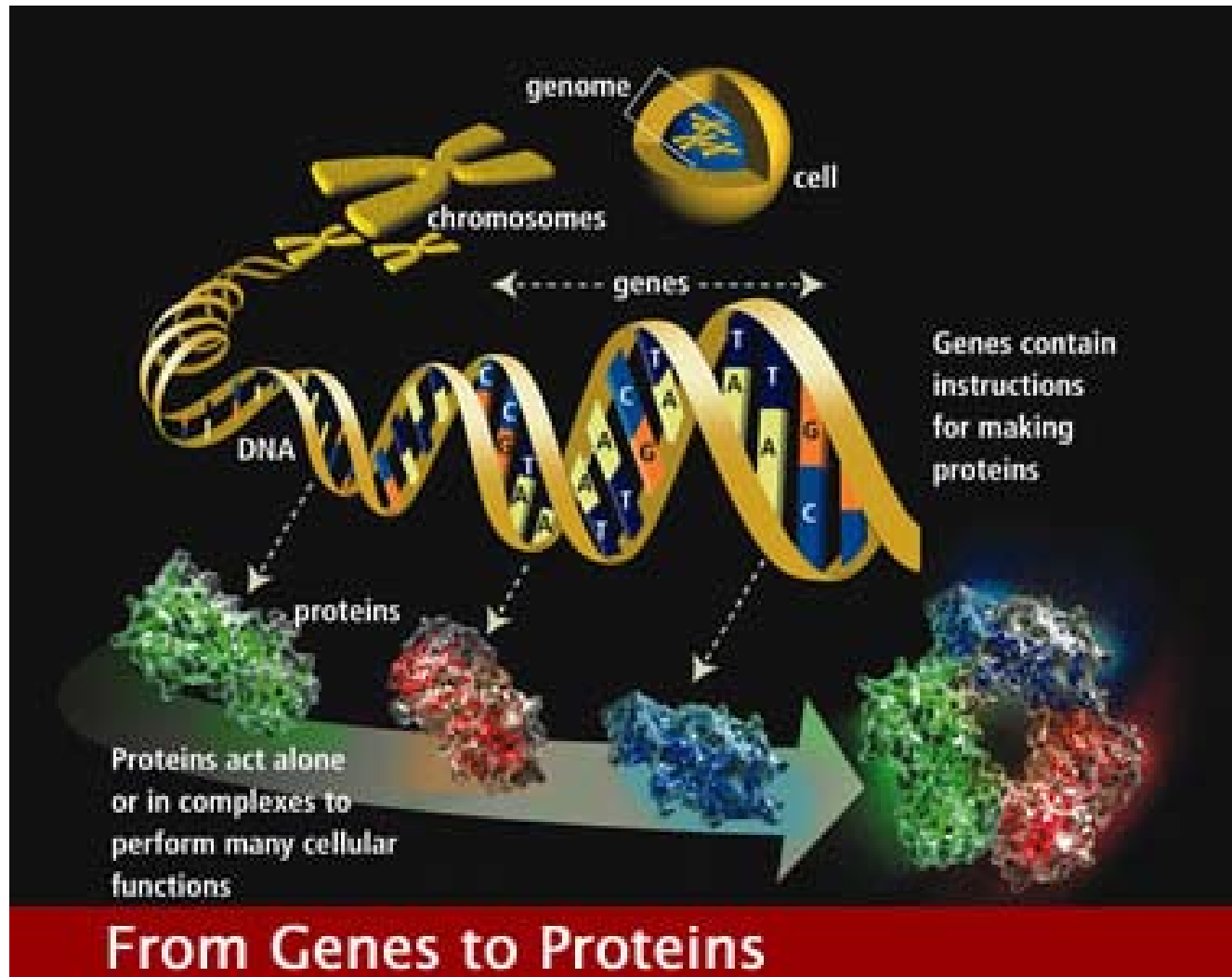
Reference Books:

1. R.C. Deonier, S. Tavaré, and M.S. Waterman,
Computational Genome Analysis, Springer. 2004
2. N. Jones and P. Pevzner,
Bioinformatics Algorithms, MIT Press. 2003
3. K.-M. Chao and L.X. Zhang,
Sequence Comparison: Theory and Methods,
Springer, 2008.

Topic 1: Sequence Comparison

- Web search tools are designed based on sequence comparison solutions.
- Text processing tools are designed based on efficient solutions to text storage and word pattern matching.
- Finally, bio-molecular sequence comparison is extremely important in modern biological and medical sciences.

1.1 Genomics Primer



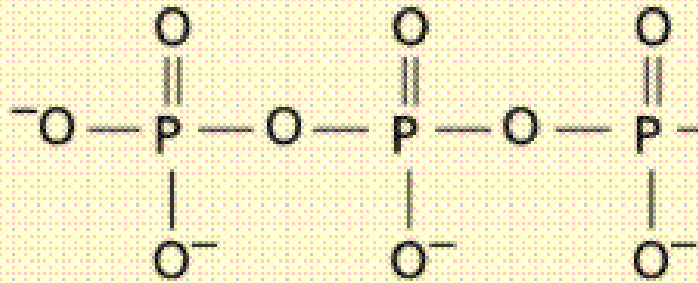
What is DNA?

- Deoxyribonucleic acids are identical for all organisms.
- Code for proteins
- the genetic material that is passed on to offsprings. Genes are the basic physical and functional units of heredity.

Molecular Structure of DNA

Deoxy-ATP
(deoxyadenosine triphosphate)

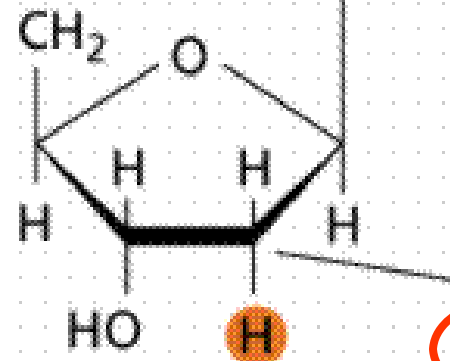
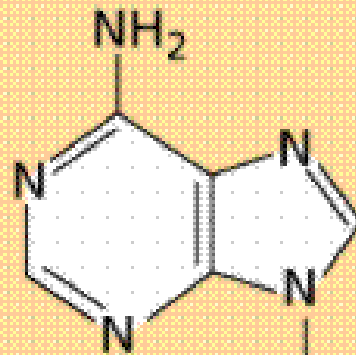
Phosphate groups



Base

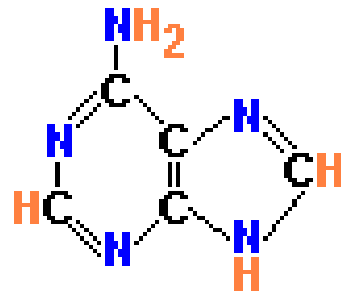
E.g.

Adenine

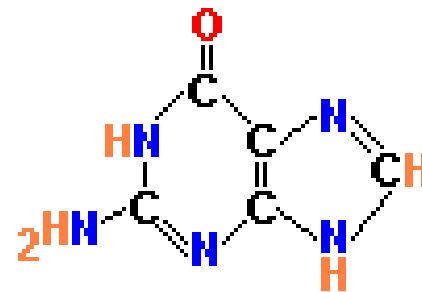


Deoxy-
ribose
sugar

Nucleic Acid Bases

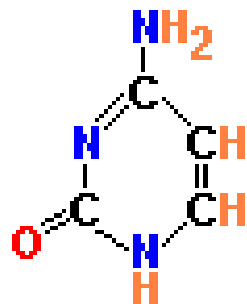


Adenine

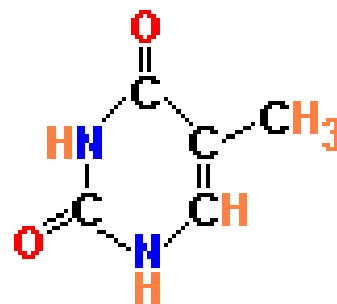


Guanine

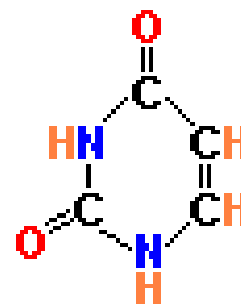
Purines



Cytosine

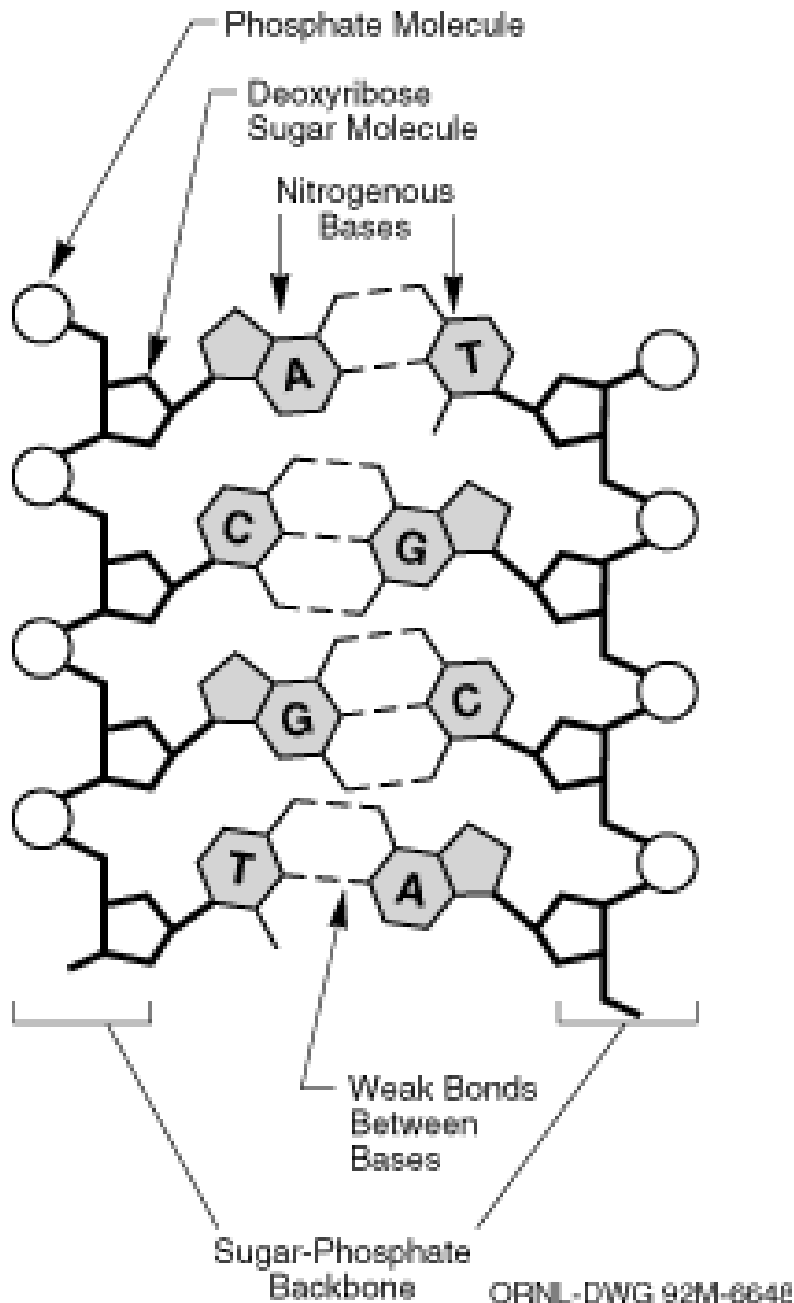


Thymine

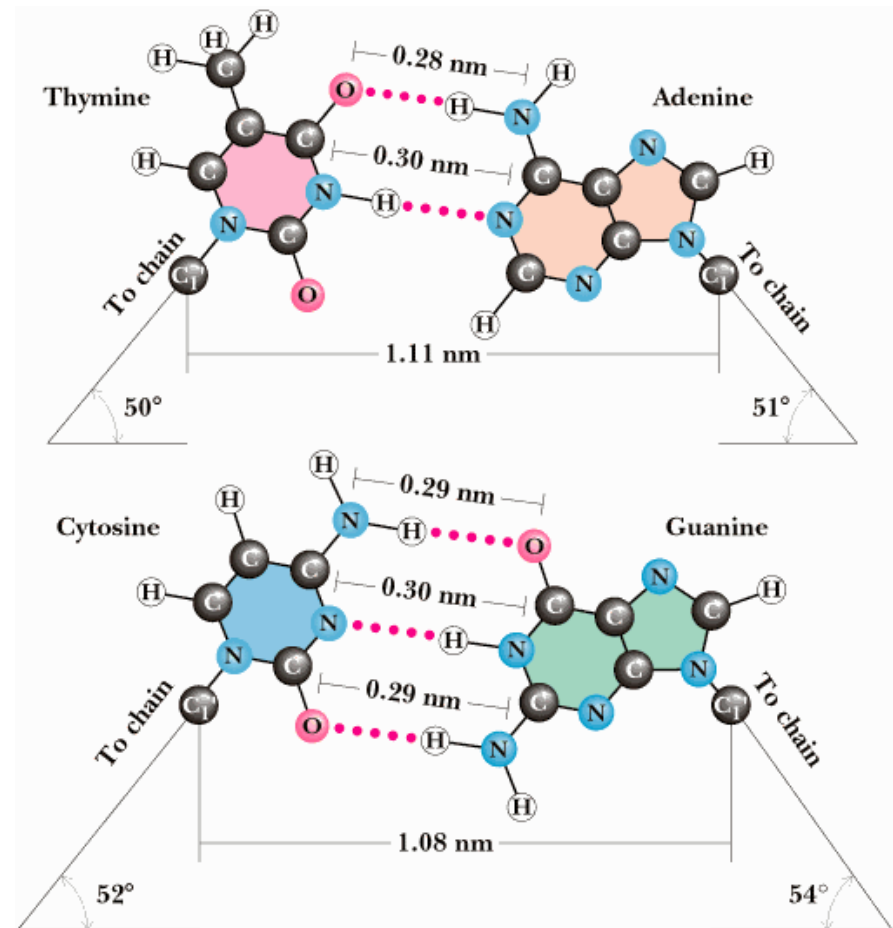


Uracil

Pyrimidines



The **DNA sequence** is the particular string of side-by-side arrangement of bases. This order spells out the exact instructions required to create a particular organism with its own unique traits.

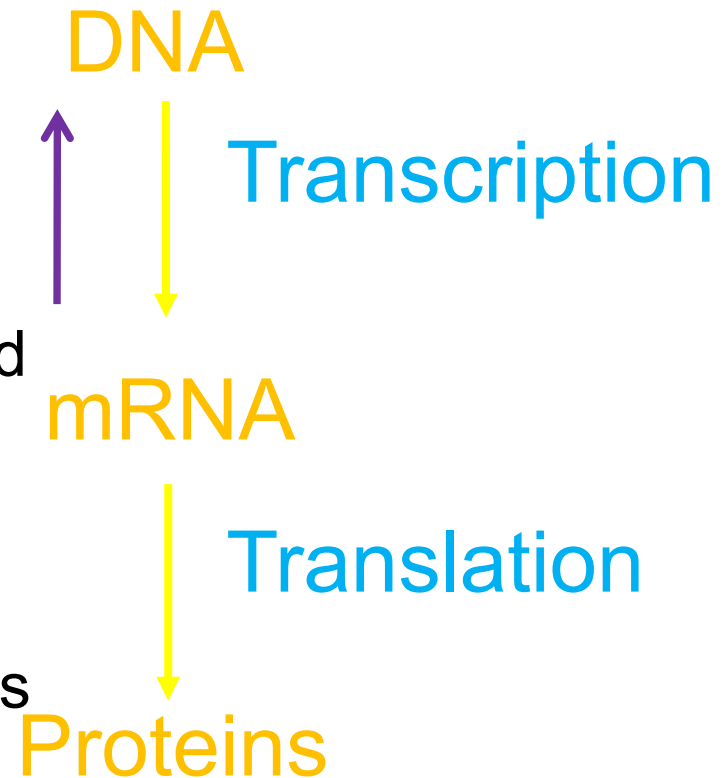


Genetic information flow

1. Transcription: a part of DNA is converted into a RNA

2. Translation: a protein is synthesized according to a RNA

3. Reverse Transcription: A RNA is used as a template for the synthesis of DNA, as a retrovirus replciation.



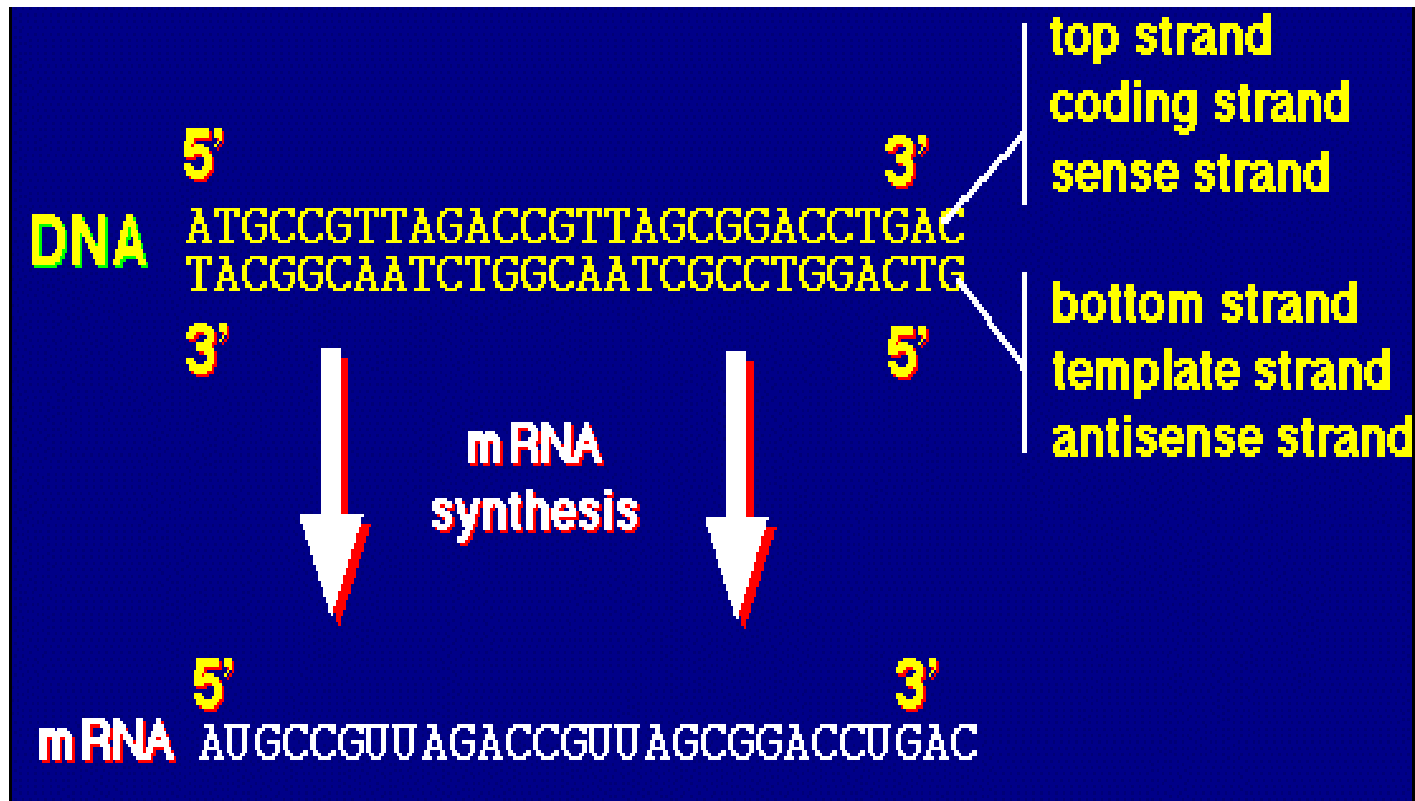
What is RNA?

- Ribonucleic acid
- usually refers to messenger RNA
- Blueprint for construction of a protein
- Actually there are 3 types RNA:
 - Messenger RNA (mRNA) Blueprint for protein
 - Ribosomal RNA (rRNA) Construction site for protein
 - Transfer RNA (tRNA) Delivery truck with designated amino acid

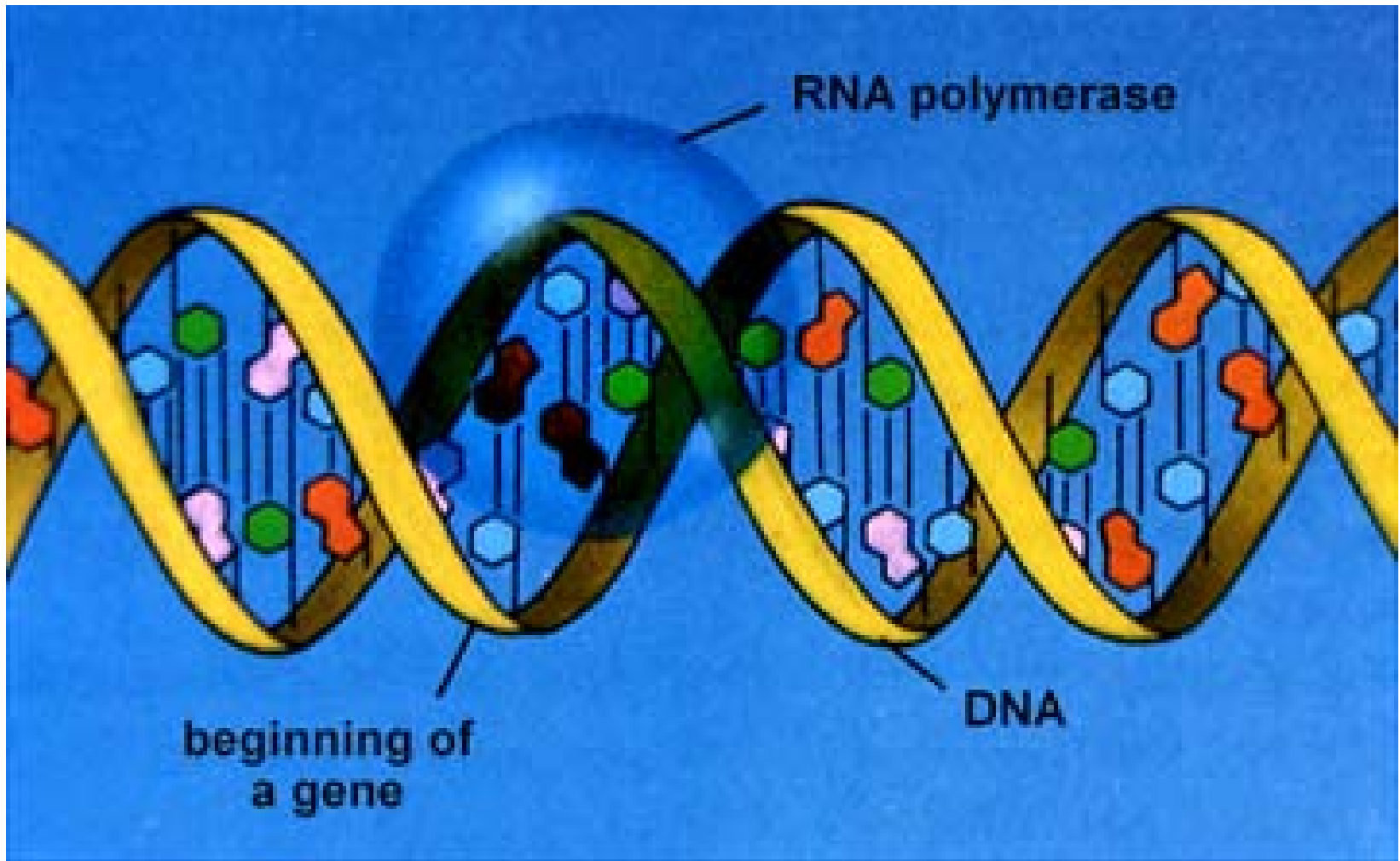
Transcription

DNA → RNA

- Making an RNA copy of a DNA sequence
- Only 1 strand of DNA is transcribed.

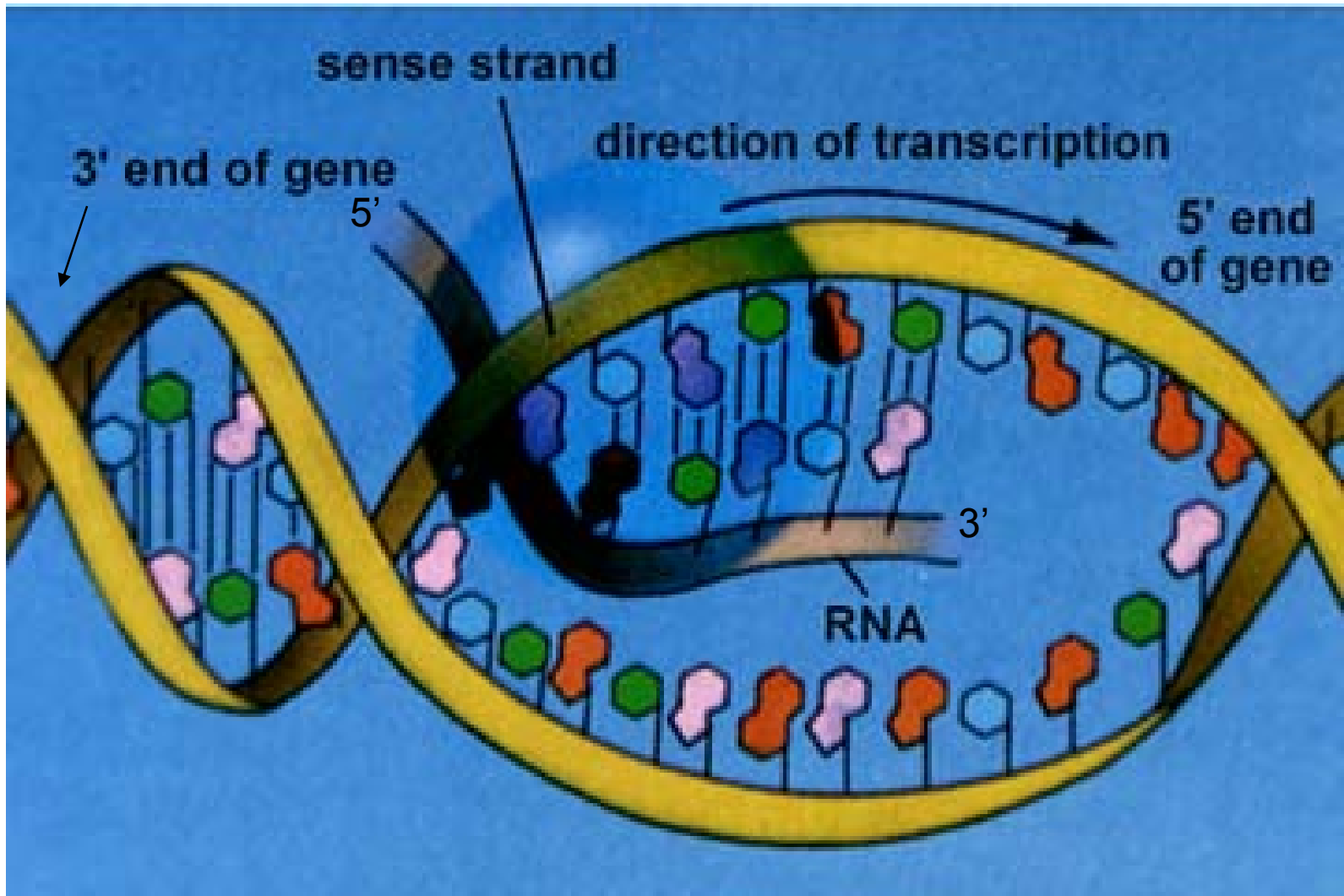


Transcription

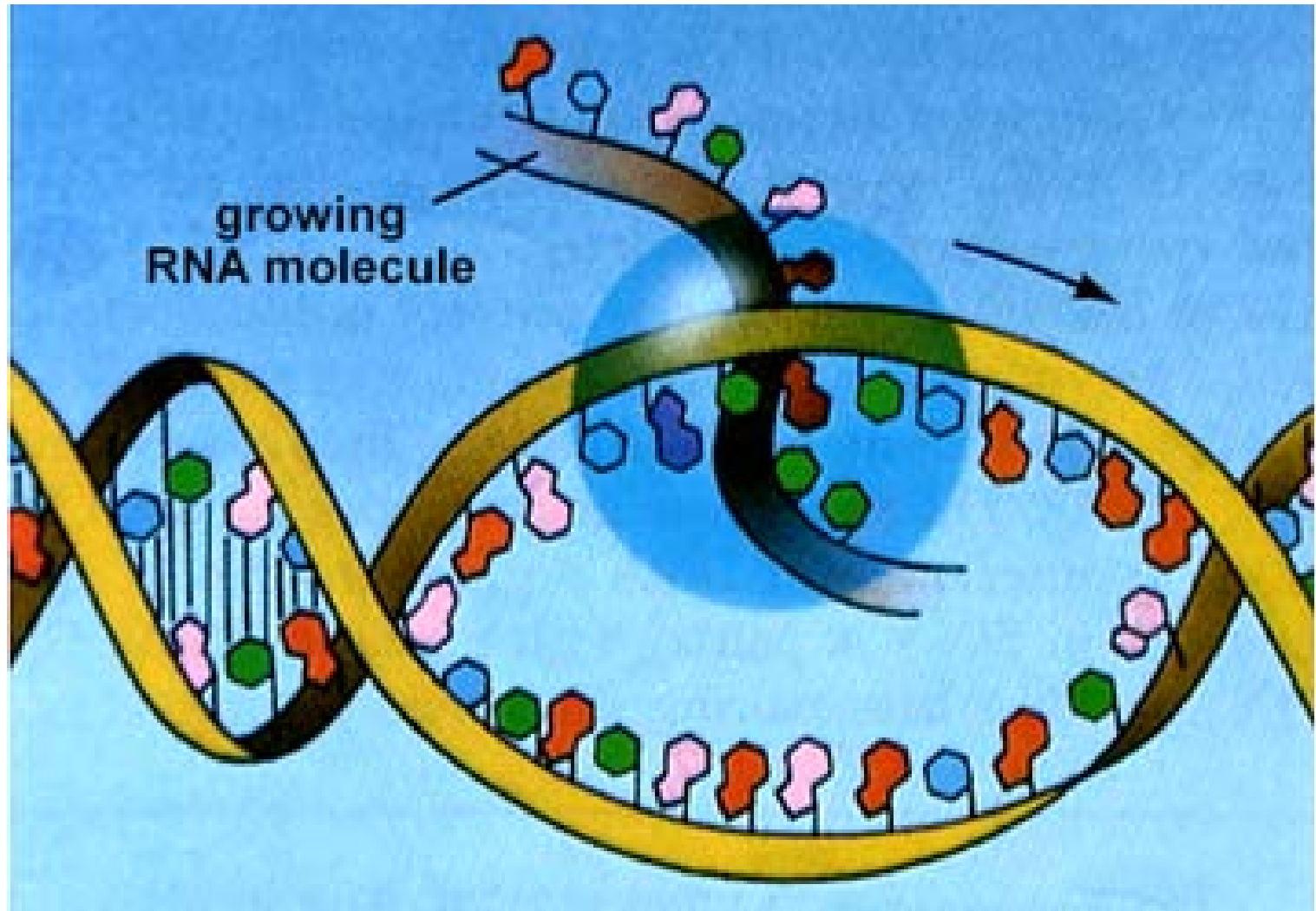


RNA polymerase opens part of DNA to be transcribed

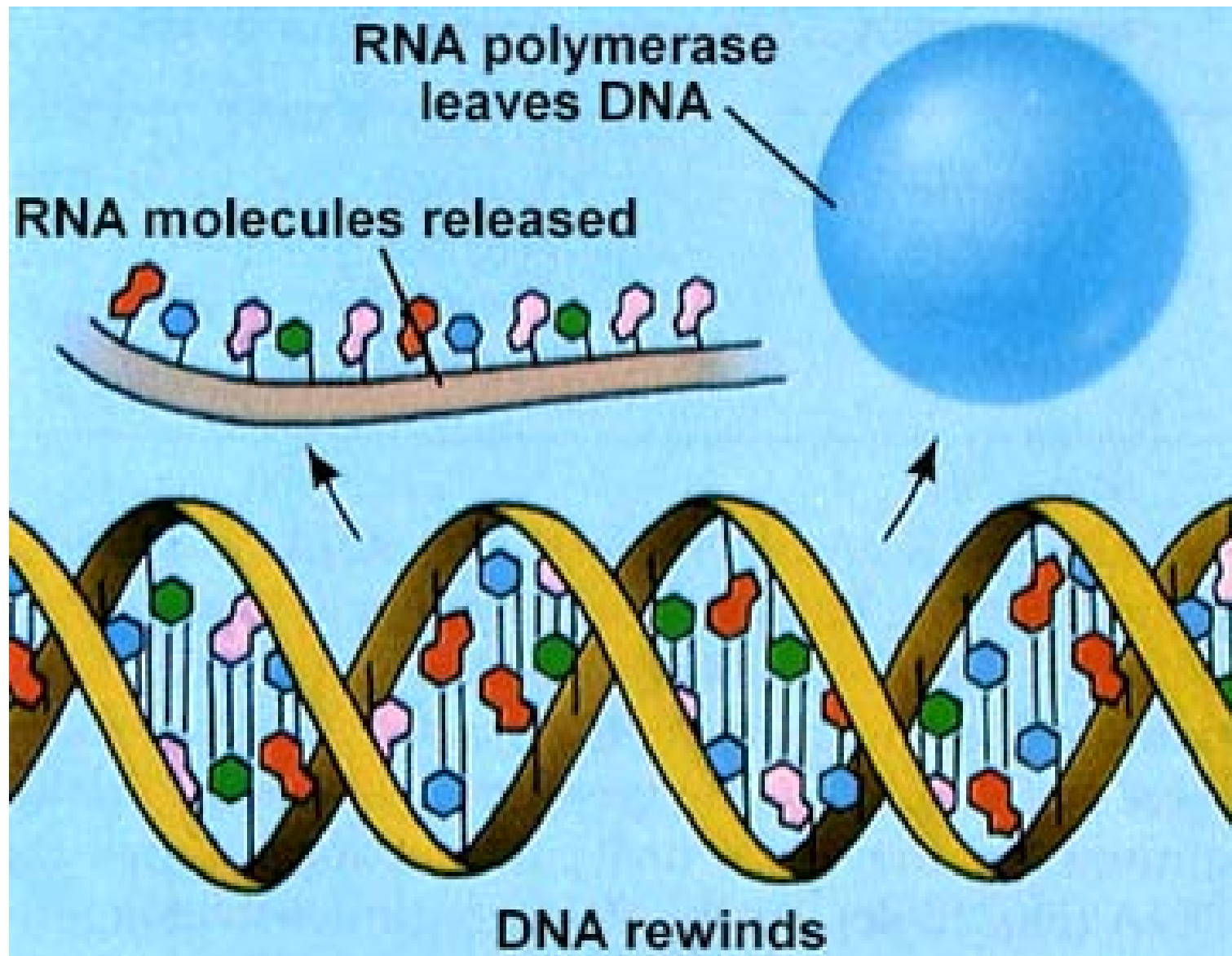
Transcription



Transcription



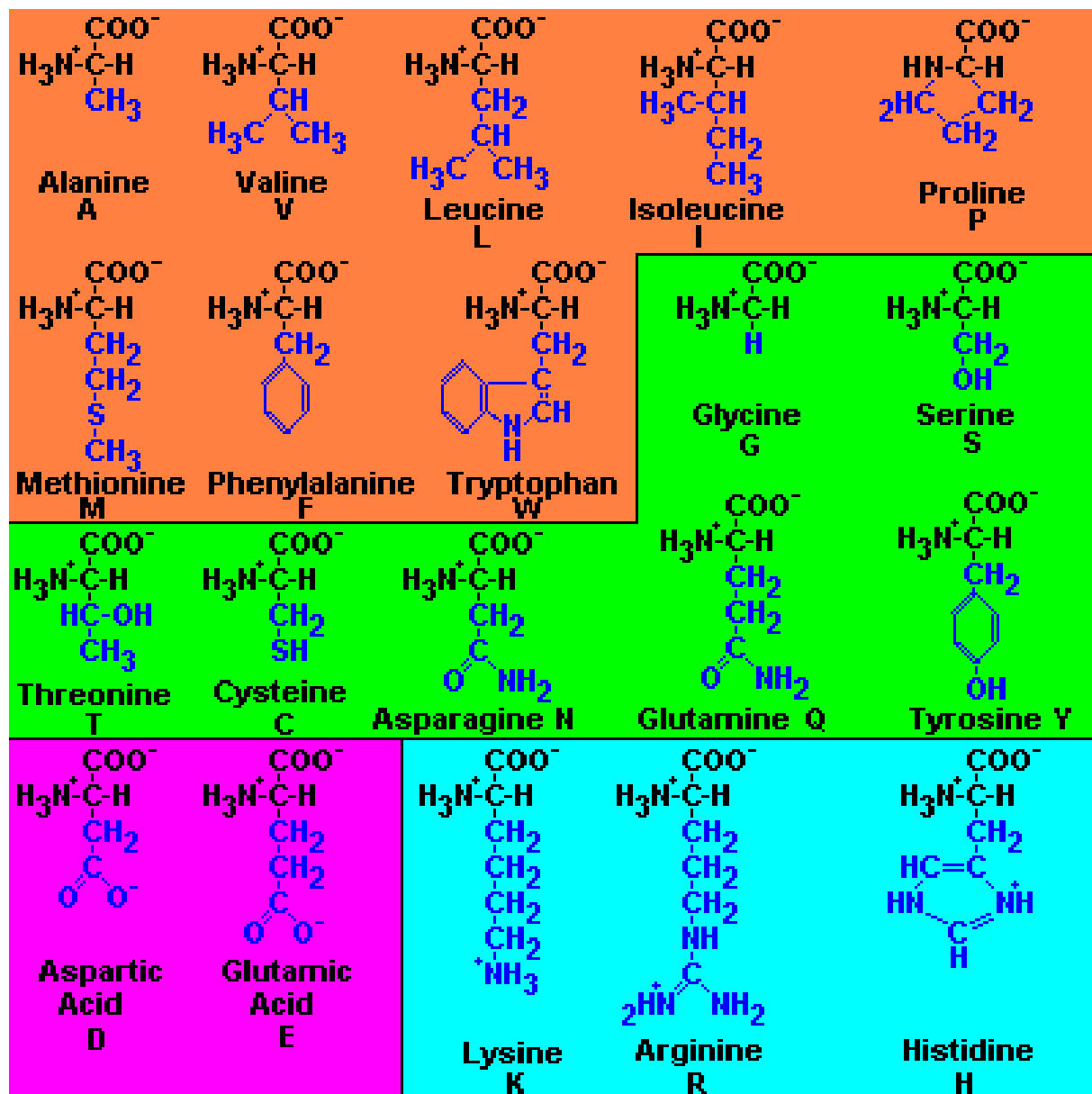
Transcription



Proteins

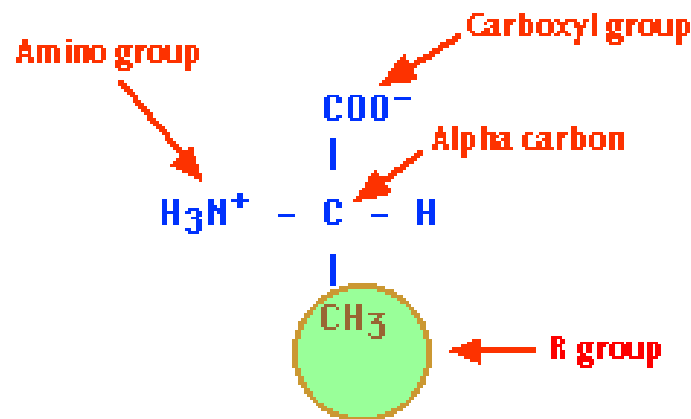
- Proteins are macromolecules and constructed from one or more strings of amino acids; that is, they are polymers.
- The **shape** and other properties of each protein depend mainly on by the precise sequence of amino acids in it.
- A typical protein contains 200-300 amino acids but some are much smaller (the smallest are often called peptides) and some much larger. The largest to date is **titin**, a protein found in skeletal muscle; it contains about 27,000 amino acids in a single chain!).

The 20 Amino Acids



Amino Acids

- Each amino acid consists of an **alpha carbon atom** to which is attached
 - a **hydrogen atom**
 - an **amino group** (hence "amino" acid)
 - a **carboxyl group** (-COOH). This gives up a proton and is thus an acid (hence amino "acid")
 - one of 20 different **"R" groups**. It is the structure of the R group that determines which of the 20 it is and its special properties.



Essential Amino Acids

- Humans must include adequate amounts of 9 amino acids in their diet. The essential amino acids for humans are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. They cannot be synthesized from other precursors. However, cysteine can partially meet the need for methionine (they both contain sulfur), and tyrosine can partially substitute for phenylalanine.
- Two of the essential amino acids, lysine and tryptophan, are poorly represented in most plant proteins.

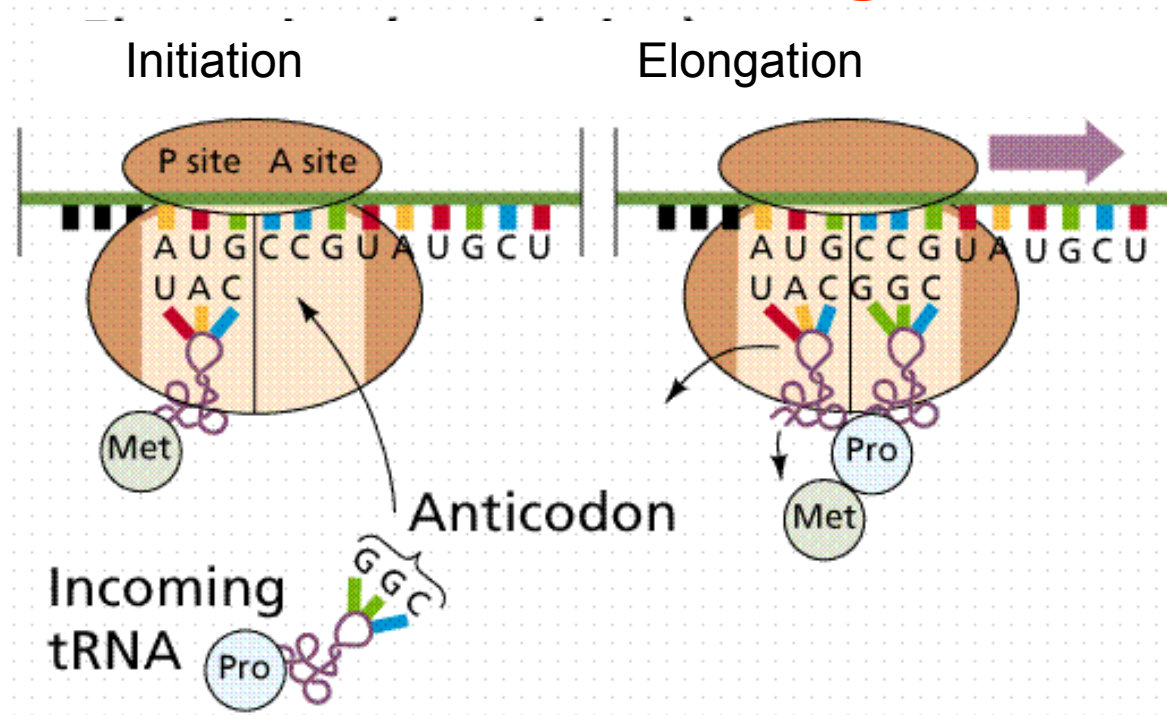
The Genetic Code

- 3 bases (triplets) of A, C, G, T are used to code for 20 essential amino acids
- 64 possible codes (4x4x4)
 - 61 amino acid coding codons
 - 3 terminating codons (“full stops”)
- 3rd base is degenerate (redundant)

		SECOND BASE					
		U	C	A	G		
FIRST BASE	U	UUU } Phe	UCU } Ser	UAU } Tyr	UGU } Cys	U	THIRD BASE
		UUC } Phe	UCC } Ser	UAC } Tyr	UGC } Cys	C	
		UUA } Leu	UCA } Ser	UAA } Stop	UGA } Stop	A	
		UUG } Leu	UCG } Ser	UAG } Stop	UGG } Trp	G	
	C	CUU } Leu	CCU } Pro	CAU } His	CGU } Arg	U	
		CUC } Leu	CCC } Pro	CAC } His	CGC } Arg	C	
		CUA } Leu	CCA } Pro	CAA } Gln	CGA } Arg	A	
		CUG } Leu	CCG } Pro	CAG } Gln	CGG } Arg	G	
	A	AUU } Ile	ACU } Thr	AAU } Asn	AGU } Ser	U	
		AUC } Ile	ACC } Thr	AAC } Asn	AGC } Ser	C	
		AUA } Met	ACA } Thr	AAA } Lys	AGA } Arg	A	
		AUG } Met	ACG } Thr	AAG } Lys	AGG } Arg	G	
	G	GUU } Val	GCU } Ala	GAU } Asp	GGU } Gly	U	
		GUC } Val	GCC } Ala	GAC } Asp	GGC } Gly	C	
		GUA } Val	GCA } Ala	GAA } Glu	GGA } Gly	A	
		GUG } Val	GCG } Ala	GAG } Glu	GGG } Gly	G	

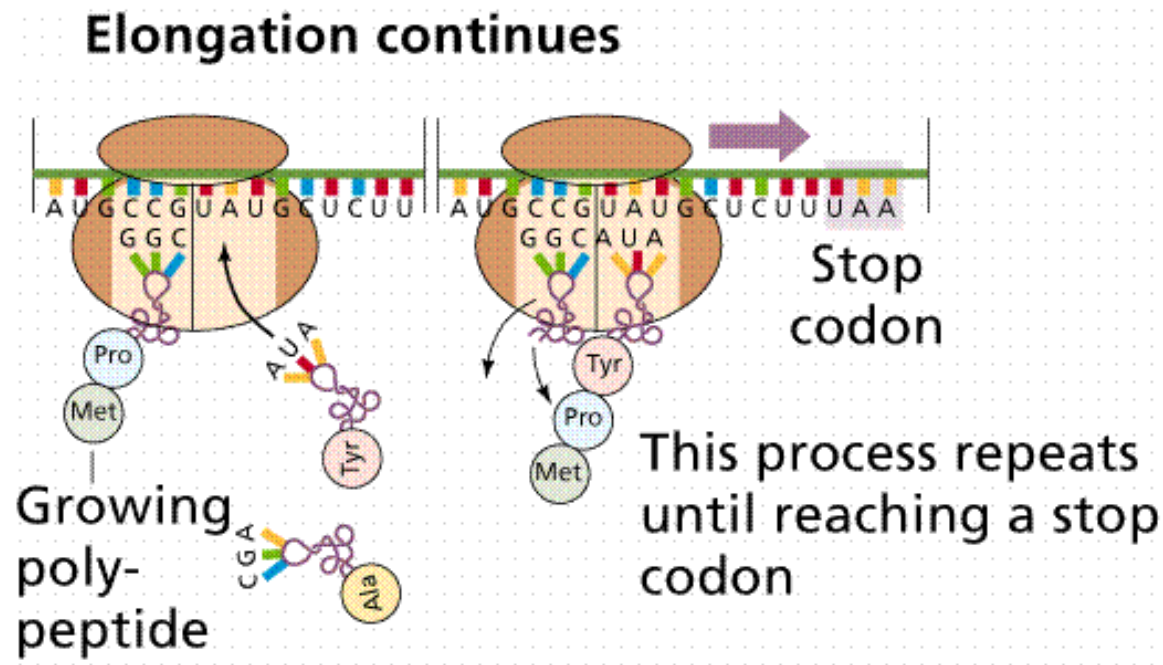
Translation (RNA to Protein)

Initiation Stage



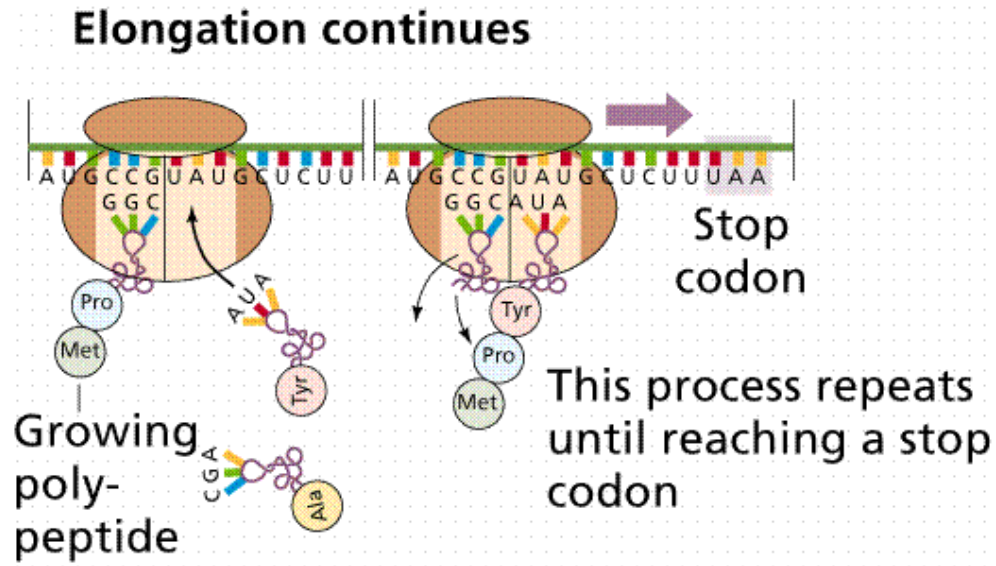
- The **small subunit** of the **ribosome** binds to a site "upstream" (on the 5' side) of the start of the message.
- It proceeds downstream (5' → 3') until it encounters the start codon AUG.
- Here it is joined by the large subunit and a special initiator tRNA.
- The initiator tRNA binds to the P site on the ribosome.
- In eukaryotes, initiator tRNA carries methionine (Met). (Bacteria use a modified methionine designated fMet.)

Translation (RNA to Protein): Elongation



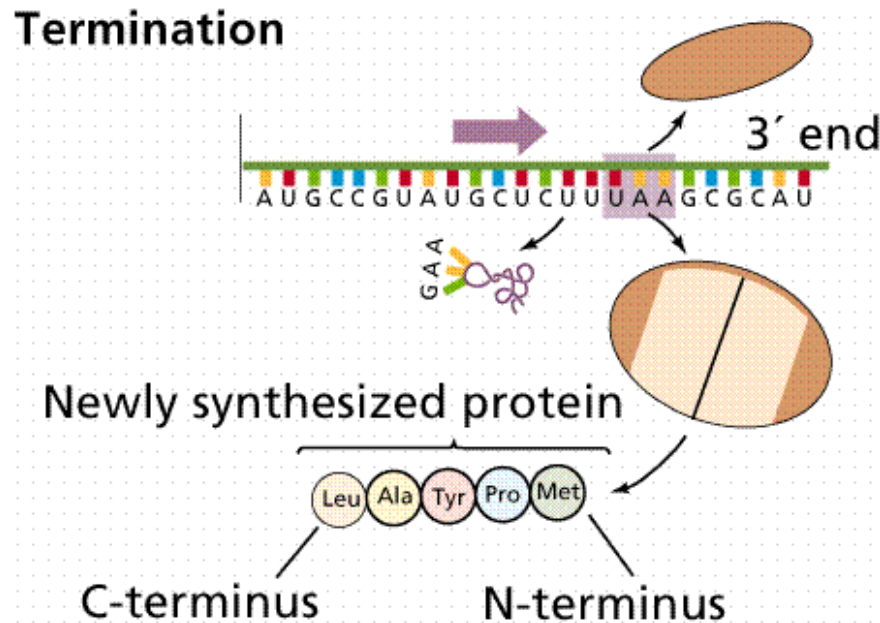
- An aminoacyl-tRNA (a tRNA covalently bound to its amino acid) able to base pair with the next codon on the mRNA arrives at the **A site** associated with:
 - an **elongation factor** (called **EF-Tu** in bacteria)
 - **GTP** (the source of the needed energy)
- The preceding amino acid (Met at the start of translation) is covalently linked to the incoming amino acid with a peptide bond.
- The initiator tRNA is released from the P site.

Translation (RNA to Protein): Elongation



- The ribosome moves one codon downstream.
- This shifts the more recently-arrived tRNA, with its attached peptide, to the P site and opens the A site for the arrival of a new aminoacyl-tRNA.
- This last step is promoted by another protein **elongation factor** (named **EF-G**) and the energy of another molecule of **GTP**.

Translation (RNA to Protein): Termination



- End of message is marked by one or more **STOP codons** (UAA, UAG, UGG).
- No tRNA molecules have anticodons for STOP codons.
- But a protein **release factor** recognizes these when they arrive at the A site.
- Binding of this protein releases the polypeptide from the ribosome.
- Ribosome splits into its subunits, can be reassembled later for another round of protein synthesis.

Genomic evolution

Whole set of DNA is the **genome** of an organism, which are arranged in to **chromosomes**.

Genomes are evolved from generation to generation.

There are **point mutations** (sub, insertion and deletion) and **segmentation mutation** (reversal, fission, fusion, etc),

-- Different genomes have different size and number of chromosomes.

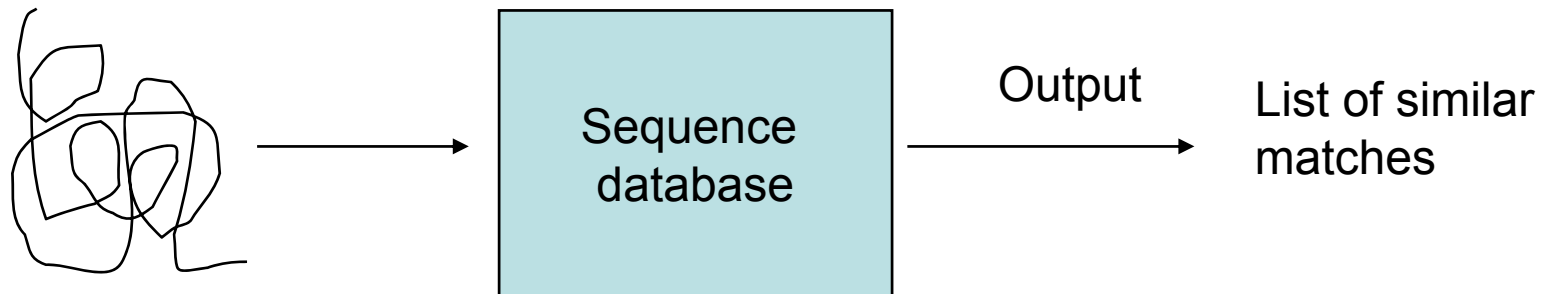
A bacterium has only as few as 600K base pairs; whereas human and mouse have 3 billions base pairs

-- Less important regions of the genome mutate or change with time more frequently than the functional important parts.

-- Hence, sequence conservation implies function. In other words, sequence conservation serves as a signal of the functionally important regions in a genome.

Applications of Sequence Comparison 1:

To determine whether the SARS is caused by a new virus or not, one may sequence the SARS virus and search against a databases of known bacterial and viral sequences.



1.2 Alignment: Model for Bio-Seq Comparison

```
Hb_a  -----VLSPADKTNVKAANGKVGA-----HAGEYGAEALERMFLSFPTTKTYFPHF
Hb_b  -----VHLIPEEKSAVTALWGKV-----NVDEVGGEALGRLLVVYPWTQRFFESF
Mb_SW -----VLSEGEWQLVLHVWAKVEA-----DVAGHGQDILIRLFKSHPETLEKFDRF
LegHb -----GALTESQAALVKSSWEEFNA-----NIPKHTHRFFILVLEIAPAAKDLFSFL
BacHb -----LDQQTINI I KATVPVLKEHG-----V-TITTFYKNLFAKHPEVRPLF---
SeaHb  GGTLAIQAQGDLT LAOKKIVRK TWHOLMR-----NKTSFVTDV FIRIFAYDPSAQNKF PQM
AscHb  -----ANKTRELCMK SLEHAKVDTSNEARQDGIDLYKHM FENYPPLRKYFKS-
Eryt.  -----LSADQISTVQASFDKVGK-----DPVGILYAVFKADPSIMAKFTQF
```

An alignment among k sequences is a k -row matrix such that

- The row j contains the j -th sequence and between two consecutive letters there may be one or more ‘-’s;
- Each column contains at least one letter.

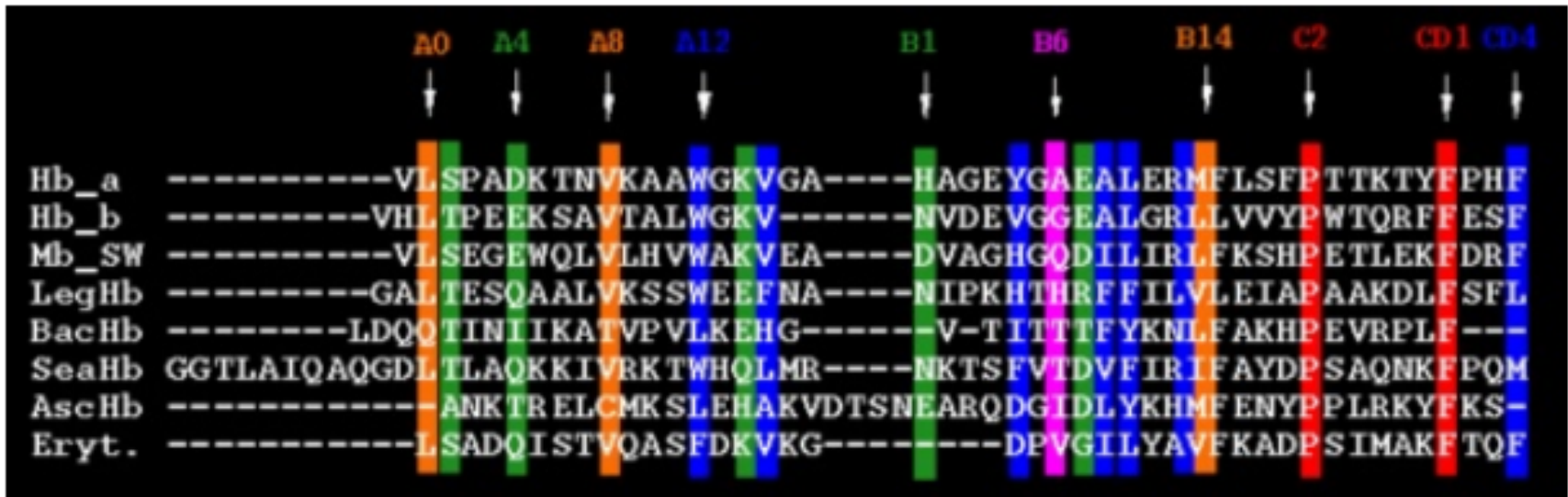
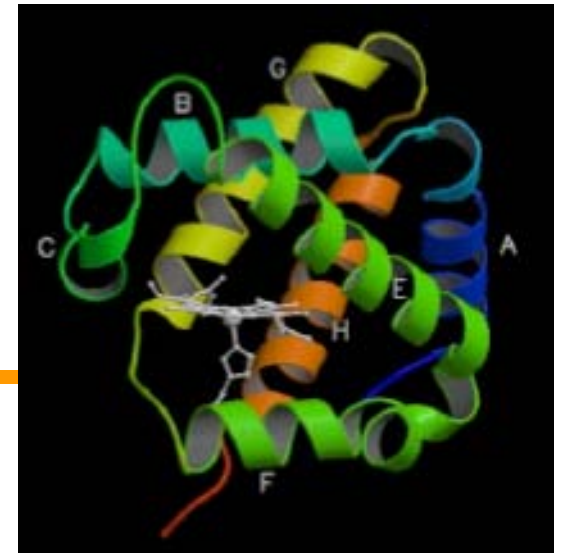
More examples

a g a a - t g c a a g a a t g - c a
a c a a g t g - a a c a a - g t g a

Each alignment is an evolutionary hypothesis, which accounts for three types of mutations:

- Substitution (a g c t → a g t t)
- Insertion (a g c t → a g a c t)
- Deletion (a g c t → a t)
- Insertions and Deletions are called **indels**.
- Consecutive dashes in a seq form a **gap**.

Alignment of globin protein sequences



Indel regions in an alignment of protein sequences in a family do not affect the structure of the proteins.

Scoring Pairwise Alignment

a	g	t	c	t	c	c
a	-	t	c	-	c	c

a	g	t	c	t	c	c
a	t	c	c	c	-	-

The quality of a pairwise alignment is measured by the sum of the column scores, one for each aligned pairs, and the scores for gaps.

The left alignment is scored by

$$s(a, a) + 3s(c, c) + s(t, t) + s(g, -) + s(t, -)$$

- Scores for aligned residues and gaps form a basic scoring scheme:

	A	G	C	T	-
A	4	-2	-1	-2	-1
G		3	-2	-1	-1
C			4	-2	-1
T				3	-1

a g t c t c c
a - t c - c c

has score

$$4 + (-1) + 3 + 4 + (-1) + 4 + 4 = 17$$

Under a meaningful scoring scheme, the score of an alignment is essentially the logarithm of likelihood ratio of the alignment between two random sequences.

Matches score a positive value;
Mismatch and indels are penalized
by scoring a negative value.

Indels and Mismatch are introduced to bring up matches that appear later.

A **sequence** is a word or string on alphabet Σ .
For DNA sequences, $\Sigma = \{a, g, c, t\}$;
For protein sequences, Σ has 20 letters;
For English words, $\Sigma = \{a, b, \dots, y, z\}$.

(Global) **Pairwise Sequence Alignment**:

Instance: Two sequences $S1$ and $S2$, and
a scoring scheme

Question: Find an alignment of $S1$ and $S2$ that
has the highest score.

Such an alignment that has the highest
alignment score is called an **optimal
alignment**.

1.3 Alignment Graph:

Compact representation of all alignments

The **length** of a sequence is the number of characters contained in it.

For example,
sequence “TATAGC” has length 6.

Let S_1 and S_2 be two sequence of length m and n respectively.

The alignment graph $A(S_1, S_2)$ of S_1 and S_2 are defined as follows:

There is one-to-one correspondence from the alignment s between S and T to the paths from left-top vertex to the right-bottom vertex.

-- Vertices are lattice points:

(i, j) , $0 \leq i \leq m$, $0 \leq j \leq n$.

In total, $(m+1)(n+1)$ vertices.

-- There is an arc from (i, j) to (i', j') if and only if

$$0 \leq i' - i \leq 1$$

and

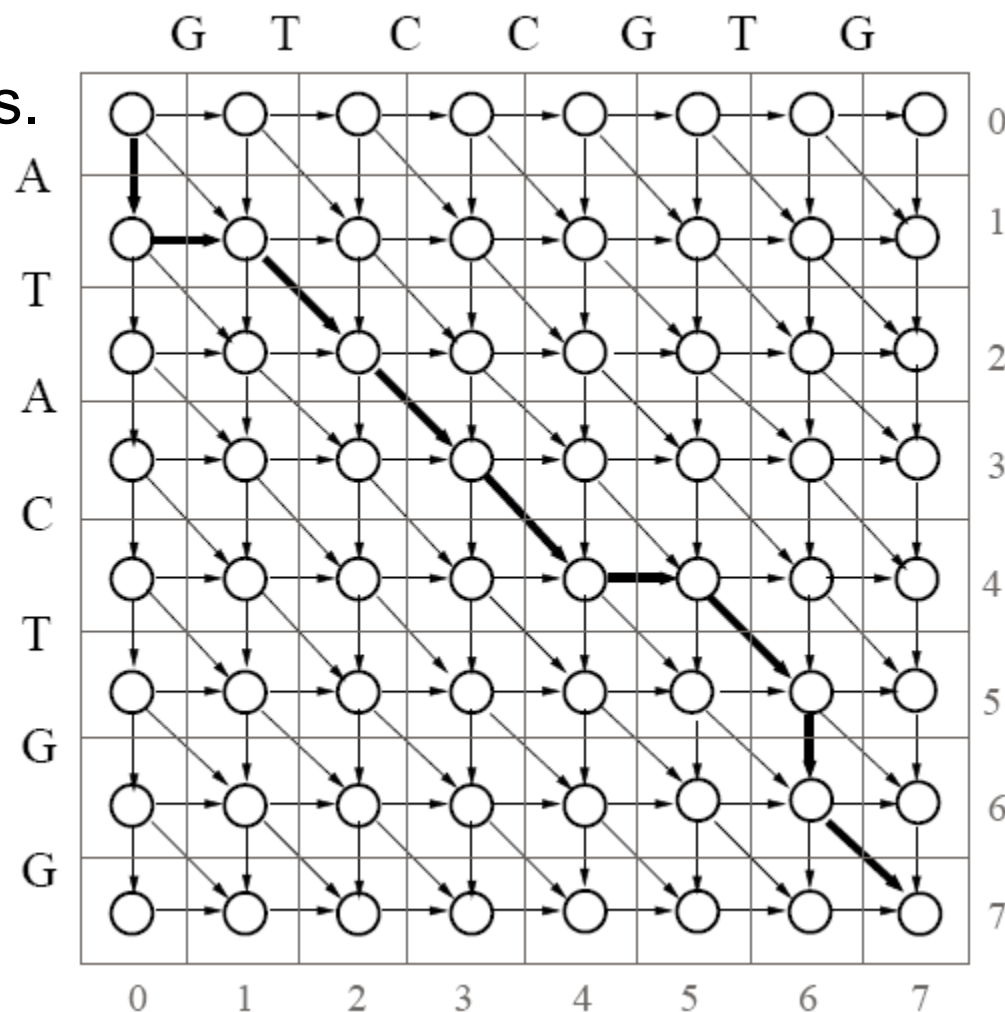
$$0 \leq j' - j \leq 1.$$

-- Three types of edges:

horizontal edges

vertical edges

diagonal edges



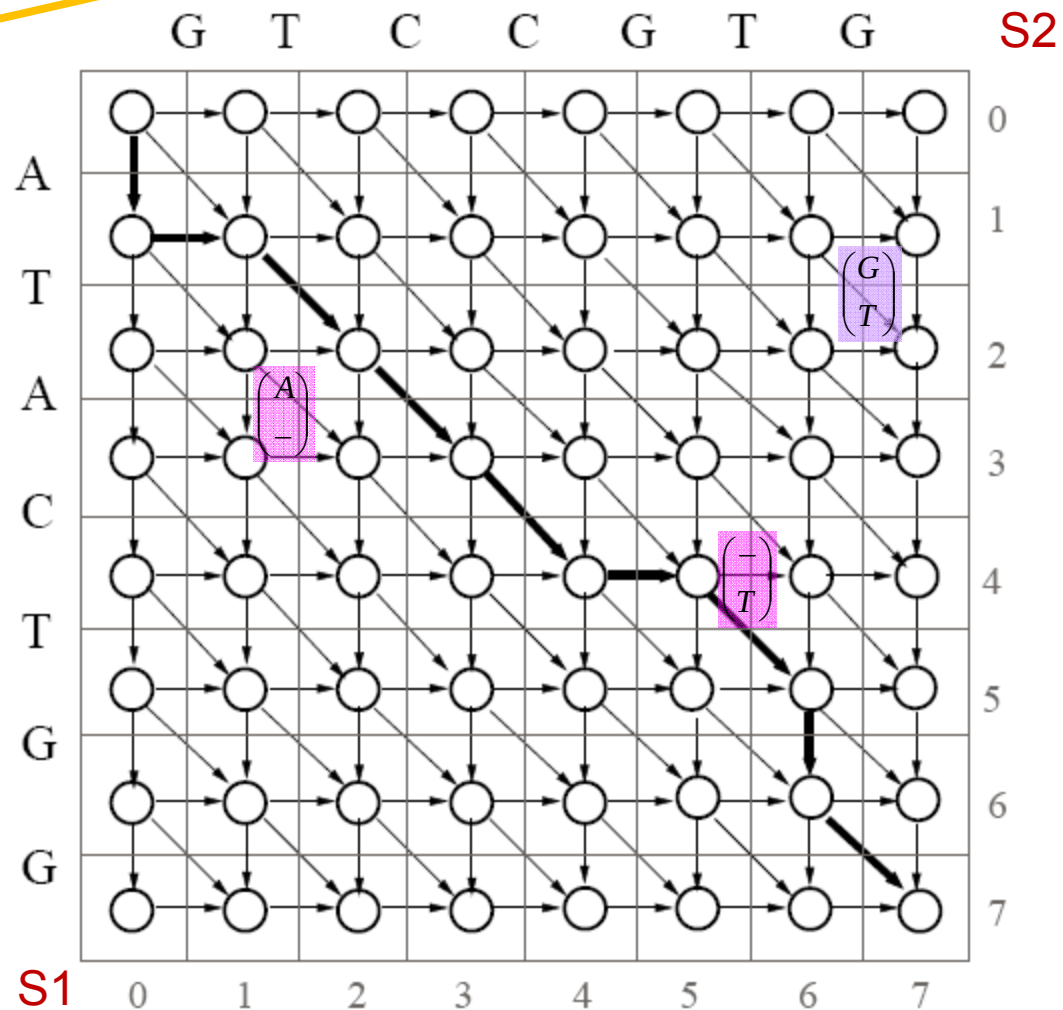
$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (3, 3) \rightarrow (4, 4) \rightarrow (4, 5) \rightarrow (5, 6) \rightarrow (6, 6) \rightarrow (7, 7)$

S1: A - T A C - T G G

S2: - G T C C G T - G

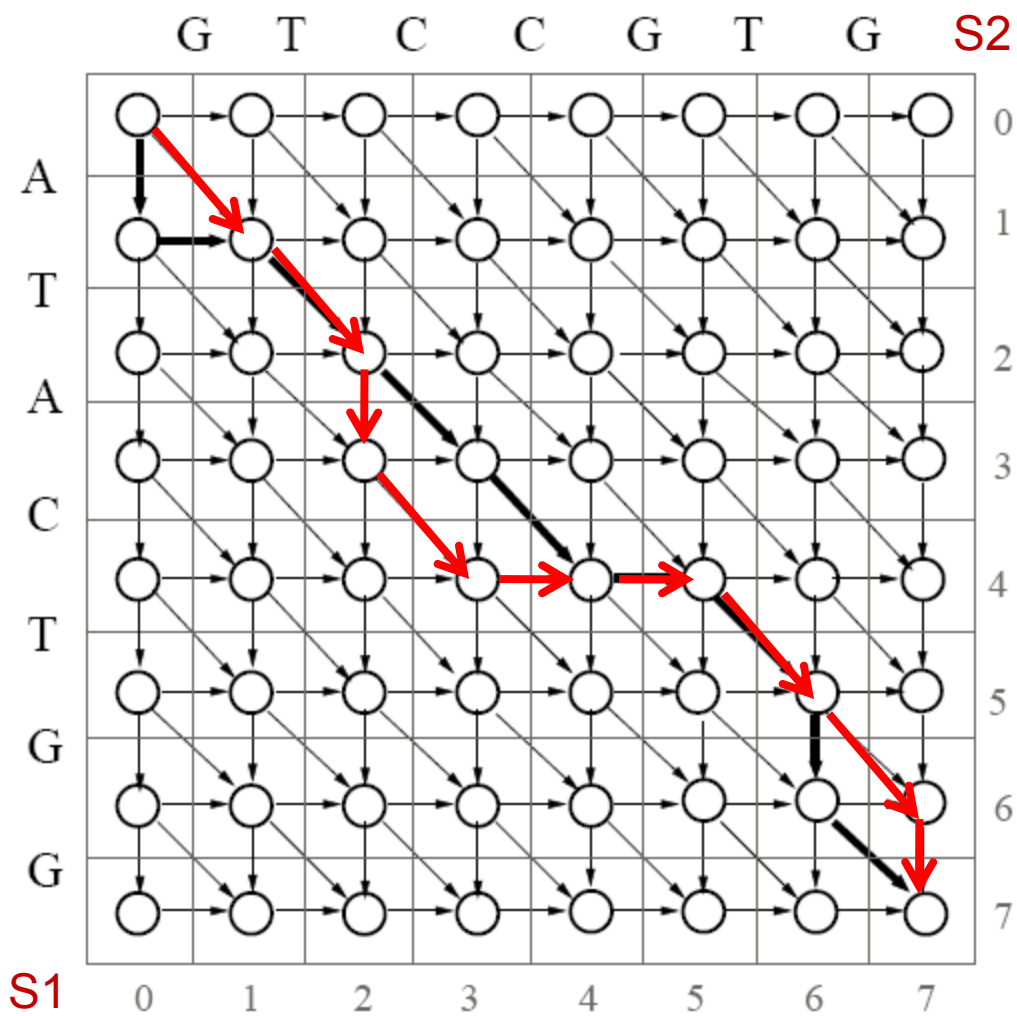
In the alignment graph, each edge corresponds uniquely to a possible column in an alignment.

- diagonal edges for match/mismatches
- horizontal and vertical edges for indels.



A T A C - - T G G

G T - C C G T G -



1.4 Generality of Alignment as a Model

Many different problems such as

- Maximum subsequence problem
- Minimum supersequence problem
- Best fitting problem
- Finding Levenshtein distance

in sequence comparison can be solved as a special case of alignment by using a specific scoring scheme.

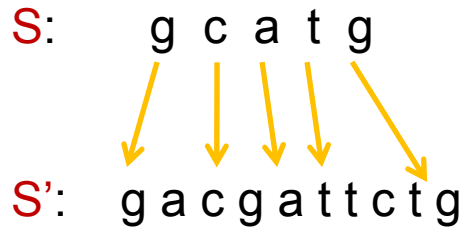
Maximum Subsequence Problem

Let S and S' be two sequences on Σ .

S is a **subsequence** of S'

(or S' is a **supersequence** of S) if

all the characters in S appear in S' in the same order.



Maximum Subsequence Problem:

Instance: Two sequences S1 and S2;

Question: Find the longest common subsequence of S1 and S2.

Diagram illustrating a partial alignment between two sequences S1 and S2 using orange lines:

S1: a g g c c a a t
S2: a c g g c t c a

Orange lines connect the following characters:

- S1[0] (a) to S2[0] (a) via a vertical line.
- S1[4] (c) to S2[4] (c) via a vertical line.
- S1[2] (g) to S2[2] (g) via a diagonal line.
- S1[5] (a) to S2[5] (t) via a diagonal line.

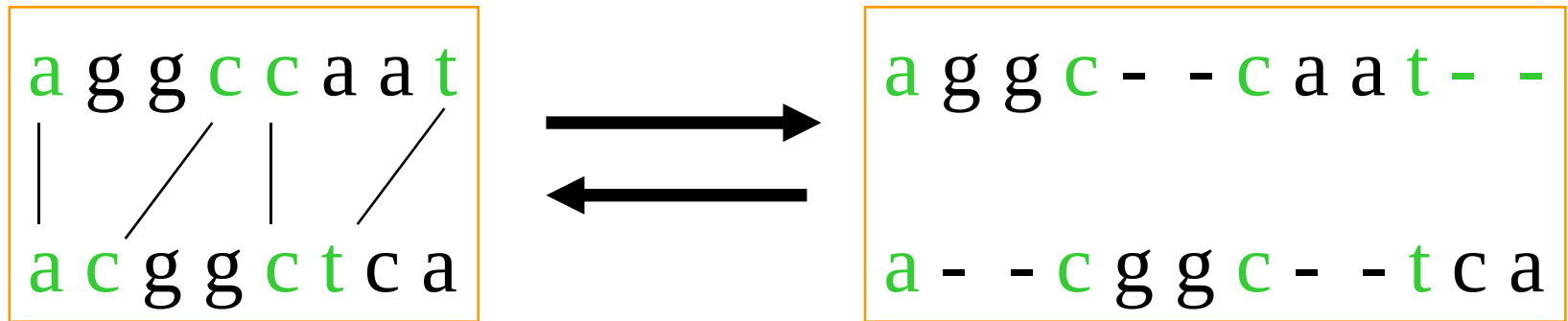
Diagram illustrating a full alignment between two sequences S1 and S2 using green lines:

S1: a g g c c a a t
S2: a c g g c t c a

Green lines connect the following characters:

- S1[0] (a) to S2[0] (a) via a vertical line.
- S1[1] (g) to S2[2] (g) via a diagonal line.
- S1[2] (g) to S2[3] (g) via a diagonal line.
- S1[3] (c) to S2[4] (c) via a diagonal line.
- S1[4] (c) to S2[5] (t) via a diagonal line.
- S1[5] (a) to S2[6] (c) via a diagonal line.
- S1[6] (a) to S2[7] (a) via a vertical line.

One-to-one correspondence



$$\downarrow \quad s(x,x)=1, s(x, y)=0$$

The length of lcs = the alignment score

The LCS problem is a special case of the alignment problem.

Levenshtein distance

In computer science and information theory, the **Levenshtein distance** between two strings is defined to be the minimum number of edit operations needed to transform one string to the other. It is also called the **edit distance**.

kitten → sitten → sittin → sitting

k	i	t	t	e	n	-
s	i	t	t	i	n	g

Finding the edit distance is equivalent to aligning sequences with match scores 0 and mismatch and indel score -1

1.5 Key Issues of Alignment

1. Algorithmic aspect:

Given a scoring scheme, how to find the optimal alignment of two sequences?

2. Scoring function:

Which score schemes are good at ranking alignments?

3. Evaluation of output alignments:

Are the output alignments statistically significant?

Homology vs Similarity

- Two sequences are **homologous** if they are evolved from a common ancestor.
- **Similarity** (score) refers to a degree of the match between two sequences.

In sequence comparison, **we derive the homology relation from similarity**. Hence,

When we say two sequences are homologous, we just state what we believe. Hence, it is important to ask how often an alignment score is expected to occur between two unrelated sequences by chance.

The number of possible alignments

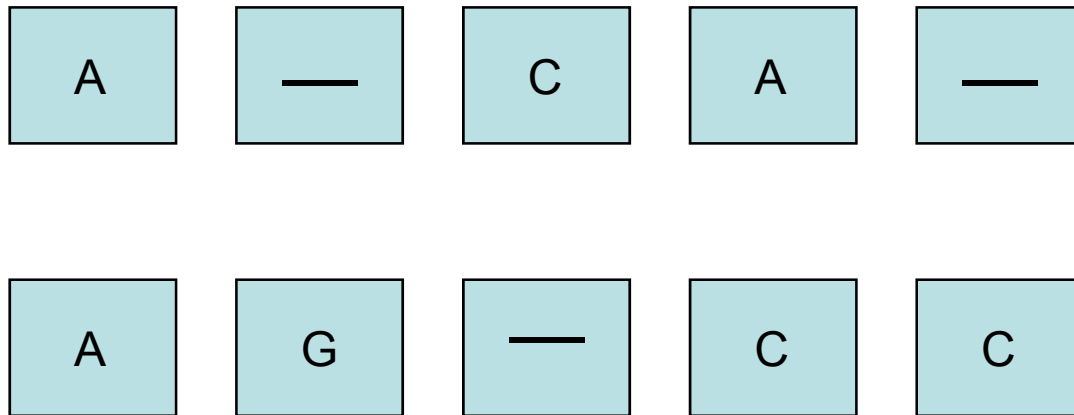
Theorem: There are

$$\sum_{i=m}^{2m} \binom{k}{m} \binom{m}{2m-k}$$

possible alignments for two m-character sequences.

Proof. An alignment of two m-character sequences has at least m columns and at most 2m columns. So we only need to prove the following fact.

Fact: There are $\binom{k}{m} \binom{m}{2m-k}$ possible alignments with k columns for two m -character sequences S and T .



Sequence S appears in the first row in $\binom{k}{m}$ ways.

After the configuration of the first row is fixed, T appears in the second row in $\binom{m}{2m-k}$ ways.